Post Midsem Learning Roadmap: AI, Computer Vision & Generative AI for Industry

From Foundations → PyTorch → Vision Tasks → GenAI → Real-world Applications (Bangalore Focus)

1. Foundations (1–2 Weeks)

- Python for AI (NumPy, Pandas, Matplotlib)
- Linear algebra basics
- Probability & statistics
- Image processing basics (OpenCV)
- Git & Dev environment (Colab / VS Code)

2. PyTorch & Deep Learning Basics (2–3 Weeks)

- PyTorch tensors & autograd
- Neural network building blocks (Linear, Conv, Pooling)
- Training loop, loss functions
- Optimization (SGD, Adam)
- GPU training
- Datasets & DataLoaders

3. CNNs for Image Classification (2–3 Weeks)

- CNN architecture (Conv, ReLU, Pooling, FC)
- Popular models (LeNet, AlexNet, ResNet, EfficientNet)
- Transfer learning
- Image augmentation
- Model evaluation (accuracy, confusion matrix)



4. Object Detection (2–3 Weeks)

- Concepts: bounding boxes, anchors, IoU, NMS
- Models: Faster R-CNN, SSD, YOLO (v5/v8), DETR
- Dataset: COCO, Pascal VOC
- Training & inference
- Evaluation: mAP
- Applications: people/vehicle detection, surveillance



5. Image Segmentation (2–3 Weeks)

- Semantic vs Instance Segmentation
- Models: U-Net, DeepLab, Mask R-CNN, Segment Anything (SAM)
- Loss functions (Dice, IoU)
- Applications: medical, remote sensing, autonomous driving, industrial inspection



6. Advanced & Generative AI (2–3 Weeks)

- Transformers & Vision Transformers (ViT)
- LLMs and multimodal models
- Generative models (GAN, Diffusion, Stable Diffusion)
- Text-to-image, image-to-image
- RAG (Retrieval Augmented Generation)



7. End-to-End Projects & Deployment (2–3 Weeks)

- Real-world projects
- Model deployment (Flask/FastAPI, Docker)
- Edge AI (Jetson, ONNX)
- MLOps (MLflow, monitoring)
- Cloud (AWS, Azure, GCP)
- Explainable AI (XAI)
- Capstone project aligned to industry need

docker aws Deployment

Key Tasks and Examples (with Visuals)

Image Classification



Dog	0.95
Cat	0.03
Car	0.01
Bird	0.01

Single label classification (e.g., species, defect type)

Object Detection



Detect and localize multiple objects (bounding boxes + class)

Semantic Segmentation



Pixel-level classification (every pixel has a class)

Instance Segmentation



Pixel-level + separate objects (e.g., each person separately)

Image Generation (GenAI)



Text-to-image generation (Stable Diffusion, DALL·E)

Vision + LLM (Multimodal)



Image understanding, VQA, defect description, report generation

Important Datasets

General Vision



ImageNet (1M+ images) | CIFAR-10/100 (60K images) | MNIST (70K images)

Object Detection



COCO (80 classes) | Pascal VOC (20 classes) | KITTI (autonomous driving)

Segmentation



Cityscapes (urban scenes) | ADE20K (indoor/outdoor) | ISIC (medical)

Domain Specific (Bangalore/India)



Traffic & pedestrians | Industrial defect datasets | Satellite & Geo-spatial | Agriculture (crop/land) | Healthcare (medical imaging)

Bangalore Industry Needs and Use Cases

IT & Product Companies
- Image/video analytics (Product features)
- Content understanding
- Retail analytics

Automotive & Autonomous (near Bangalore)
- ADAS, traffic monitoring
- Driver/vehicle detection
- Lane & road segmentation
- Sensor fusion (camera, LiDAR)

Aerospace & Defence (HAL, DRDO, ISRO)
- Satellite image analysis
- UAV/drone analytics
- Target detection
- Change detection

Manufacturing & Industrial (Whitefield, Peenya, Bidadi)
- Defect detection
- Visual inspection
- Predictive maintenance
- Safety monitoring (PPE detection)

Healthcare & MedTech (Hospitals, Startups)
- Medical image analysis (CT, MRI, X-ray)
- Segmentation & diagnosis support
- AI for pathology/microscopy

Agriculture & Environment (Agri-tech, Forest, Water)
- Crop monitoring
- Land use classification
- Forest fire detection
- Environmental monitoring

Smart Cities & Surveillance (Bangalore city projects)
- Traffic flow analysis
- Crowd counting
- CCTV analytics
- Smart infrastructure

Generative AI & Enterprise (IT services, Startups)
- Text-to-image, video generation
- Document analysis (OCR + LLM)
- RAG systems for enterprise data
- AI agents and workflow automation

Tools & Frameworks to Learn

Python | NumPy | Pandas | Matplotlib | OpenCV | PyTorch | TorchVision | Hugging Face | Ultralytics (YOLO) | Detectron2 | MMDetection | Segmentation Models (PyTorch) | LangChain | Streamlit | Docker | AWS/GCP | MLflow

DIP to Modern AI: Important Acronyms, Full Forms, Uses and Placement Priority

Must Know (★★★★★)
Essential for most CV/AI jobs (Core Skills)

Acronym	Full Form	What it is / Key Use		Why Important for Placement	Example / Application
	Python	Python Programming Language	General purpose language for AI/ML/CV	Used in almost every AI job	Data processing, model development
	OpenCV	Open Source Computer Vision Library	Image/video processing library	Must know for CV roles	Filtering, edge detection, object detection, video analysis
	PyTorch	Python Torch (Deep Learning Framework)	Build ad train deep learning models	Industry standard framework	CNN, YOLO, U-Net, Transformers
	CNN	Convolutione Neural Network	Deep learning model for image feature extraction	Core concept in all CV tasks	Image classification, detection, segmentation
YOLO	YOLO	You Only Look Once	Real-time object detection model	Most demanded detection model	Person/vehicle detection, surveillance, industrial inspection
	FastAPI	Fast Application Programming Interface	Python framework for model serving (REST API)	Used to deploy AI models	Web service for image upload and prediction
	Docker	Docker Container Platform	Package application with dependencies	Required for deployment	Run AI app anywhere (laptop/ cloud)
	Git	Version Control System	Track code changes	Mandatory for collaborative work	Manage projects, resume repository
aws	AWS	Amazon Web Services	Cloud platform for hosting and scaling AI applications	Most widely used cloud platform	Deploy model on EC2/SageMaker, store data in S3

Good to Know (★★★★)
Strongly useful (Enhances CV and Job Opportunities)

Acronym	Full Form	What it is / Key Use	Why Important for Placement	Example / Application
U-Net	U-Net	U-shaped Network	Deep learning model for semantic segmentation	Tumor segmentation, land cover mapping
	SAM	Segment Anything Model	Foundation model for promptable segmentation	Interasingly used in industry
	ViT	Vision Transformer	Transformer-based model for images	Modern architecture, used in many advanced CV models
	RAG	Retrieval-Augmented Generation	Combines LLM with external knowledge	Important for vision + language applications
	ONNX	Open Neural Network Exchange	Model format for cross-platform deployment	Export Pyeloy models across diffeent frameworks
mlflow	MLflow	Machine Learning Lifecycle Platform	Experiment tracking, model management	Useful for MLOps and production

Advanced (★★★)
Good for specialized roles (Research, Advanced CV/GenAI)

Acronym	Full Form	What it is / Key Use	Why Important for Placement	Example / Application
	DETR	Transformer-based object detection model	Modern approach, used in advanced CV systems	End-to-end object detection
	DeepLab	Advanced semantic segmentation model	Used in research and specialized industry applications	Land cover mapping, urban studies
	Stable Diffusion	Latent Diffusion Model	Popular in creative AI, design and simulation	Text-to-image generation
	GAN	Generative Adversarial Network	Useful for data generation, enhancement	Image synthesis, super-resolution
	XAI	Explainable AI	Techniques to inetrpret and explain model decisions	Visualizing model attention, decision explanation

Other Important Acronyms (Frequently Used in For Placement)

Acronym	Full Form	What it is / Key Use	Example / Application
NumPy	Numerical Python	Fast matrix and tensor operations	Array operations, image data
Pandas	Python Data Analysis Library	Data loading and preprocessing	CSV/Excel/open data
Matplotlib	Python Plotting Library	Visualization	Plot images, graphs, metrics
Jupyter	Jupyter Notebook	Interactive development environment	Prototyping, experimentation
HOG	Histogram of Oriented Gradients	Feature descriptor for object detection	Pedestrian/vehicle detection
SIFT	Scale Invariant Feature Transform	Keypoint feature descriptor	Image matching, object recognition
SURF	Speeded Up Robust Features	Keypoint feature descriptor	Image matching
FFT	Fast Fourier Transform	Frequency domain analysis	Image filtering, compression
DWT	Discrete Wavelet Transform	Multi-resolution analysis	Image compression, denoising

Acronym	Full Form	What it is / Key Use	Example / Application
ReLU	Rectified Linear Unit	Activation function	Deep learning models
FC	Fully Connected Layer	Final classification layer	Image classification
LeNet	LeCun Network	Early CNN model	Handwritten digit recognition
AlexNet	Alex Krizhevsky Network	First deep learning breakthrough (2012)	Image classification
ResNet	Residual Network	Deep CNN with skip connections	High accuracy image recognition
EfficientNet	Efficient Chotal Network	Efficient CNN architecture	High accuracy, fewer parameters
SGD	Stochastic Gradient Descent	Optimizer	Training deep learning models
Adam	Adaptive Moment Estimation	Popular optimizer	Almost all deep learning models
COCO	Common Objects in Context	Standard detection dataset	Object detection benchmark
VOC	Pascal Visual Object Classes	Classical detection dataset	Object detection benchmark

Evaluation Metrics (Interview Important)

Acronym	Full Form	What it is Measures	Used In
IoU	Intersection over Union	Overlap between predicted and ground truth boxes/masks	Object detection, segmentation
mAP	Mean Average Precision	Detection performance metric	Object detection
NMS	Non-Maximun Suppression	Removes overlapping boxes	Object detection
Dice Loss	Dice Similarity Loss	Overlap metric for segmentation	Medical image segmentation
IoU Loss	Intersection over Union Loss	Loss function for segmentation	Image segmentation

End-to-End Pipeline: From Image to Deployed AI Application

1. Image Acquisition
- Camera / Drone / CCTV / Satellite
- Real-world images

2. Preprocessing (OpenCV)
- Resize, filter, enhance
- Histogram, normalization
- Morphology

3. Model Development (PyTorch)
- CNN / YOLO / U-Net / ViT
- Train, validate, test

4. Model Serving (FastAPI)
- Upload image
- Get prediction (JSON)

5. Containerization (Docker)
- Package model + dependencies

6. Cloud Deployment (AWS/Azure/GCP)
- Deploy on cloud server
- Scale, monitor, secure

7. End Users (Web/Mobile)
- Real-world application (e.g., surveillance, inspection, medical analysis, agriculture)

Placement Priority Legend

- **Must Know (★★★★★)** — Essential for most CV/AI jobs (Highest priority)
  - Python, OpenCV, PyTorch, CNN, YOLO,
  - FastAPI, Docker, Git, AWS basics
- **Good to Know (★★★★)** — Strongly useful (Enhances CV and job opportunities)
  - U-Net, SAM, ViT, RAG, ONNX, MLflow
- **Advanced (★★★)** — Good for specialized roles (Research, Advanced CV/GenAI)
  - DETR, DeepLab, Stable Diffiusion, GAN, XAI